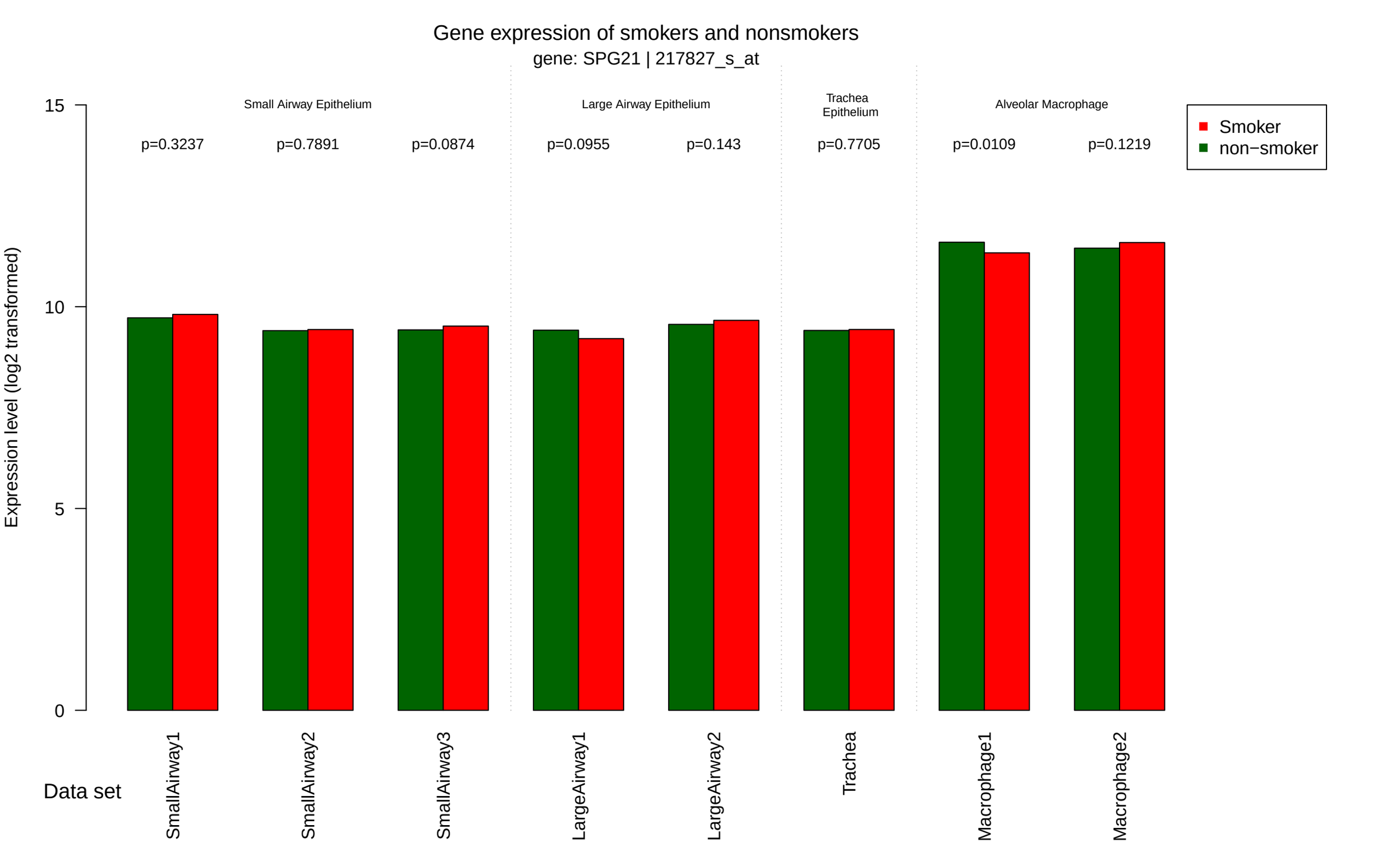
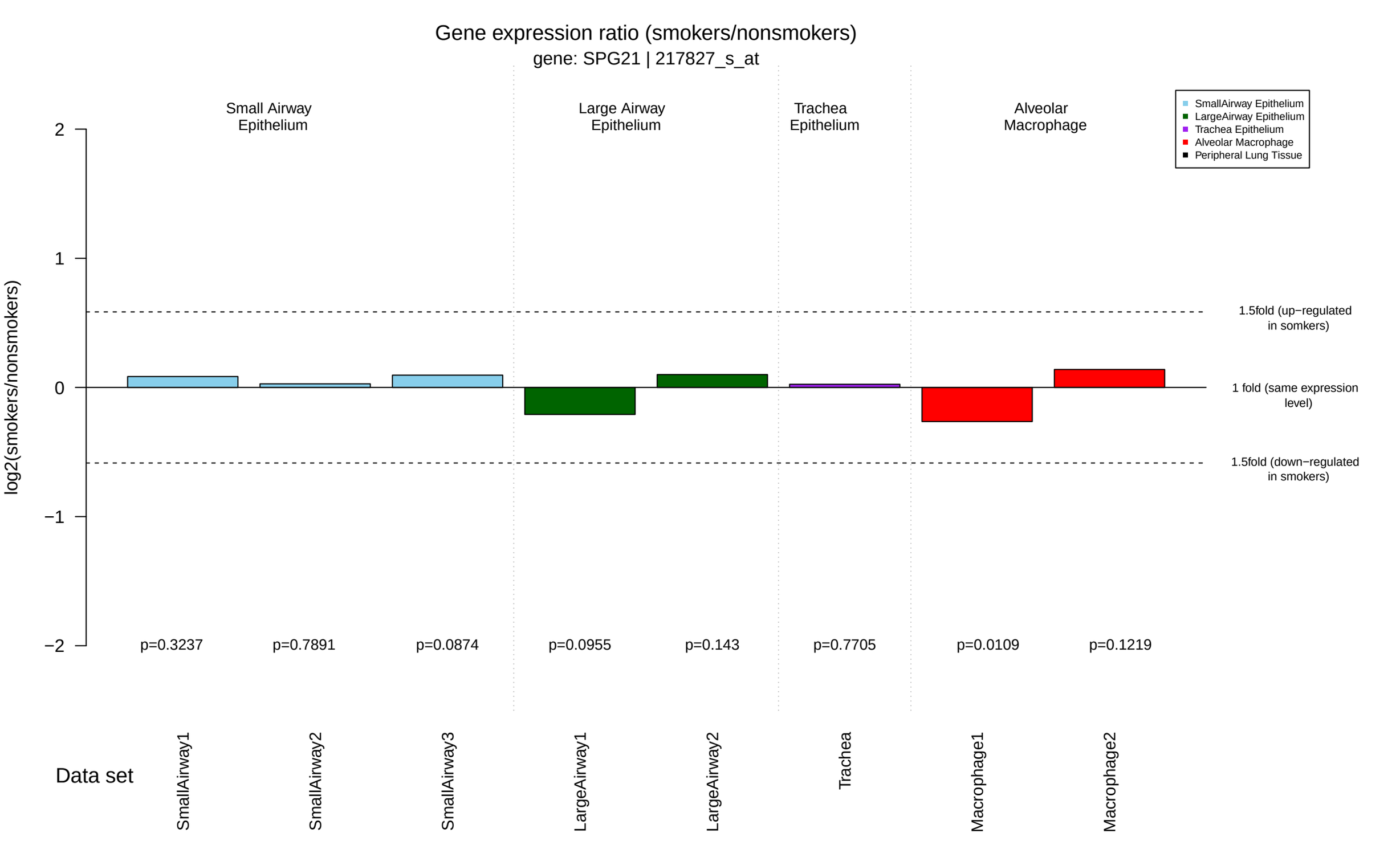
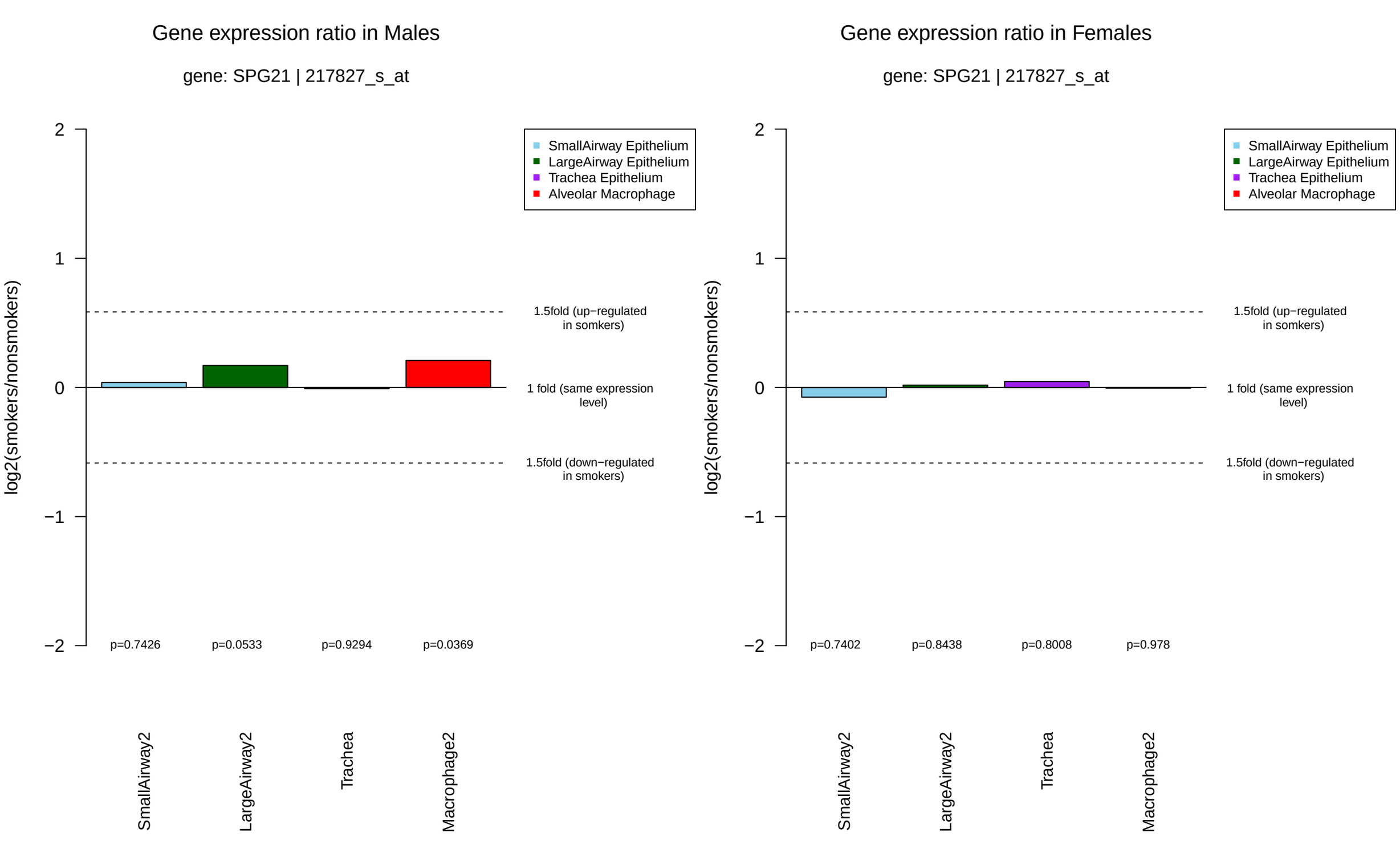
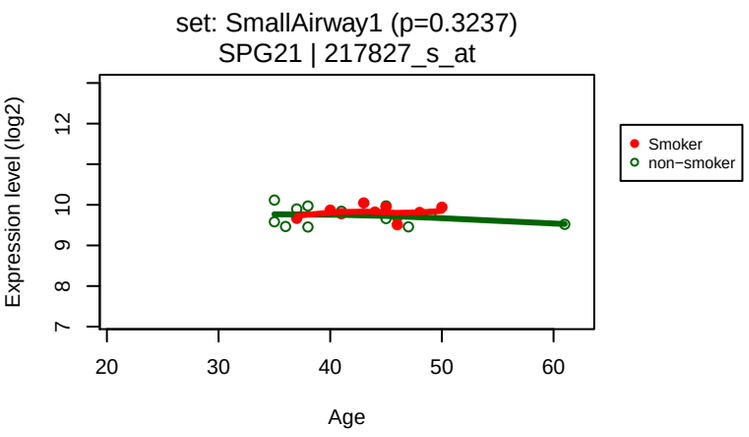
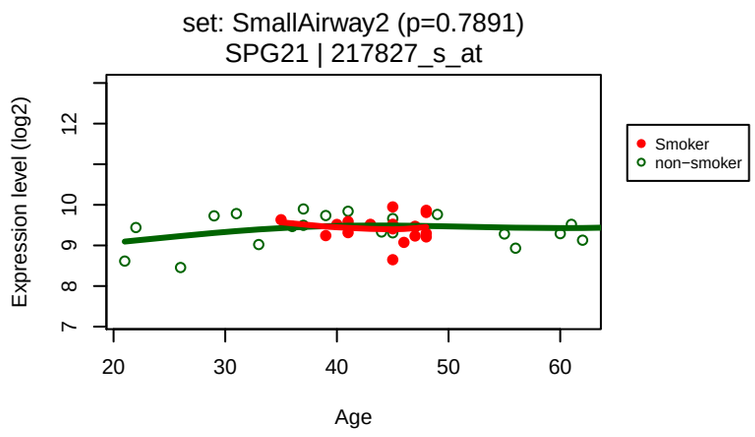
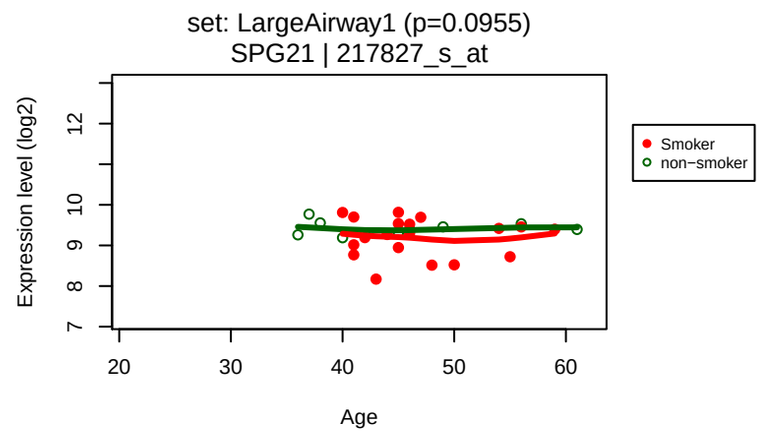

Gene expression of smokers and nonsmokers
gene: SPG21 | 217827_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.3237
p=0.7891
p=0.0874
p=0.0955
p=0.143
p=0.7705
p=0.0109
p=0.1219
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

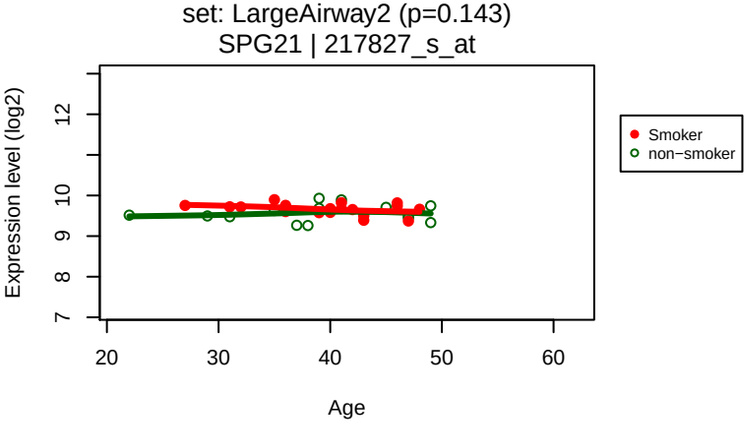

Gene expression ratio (smokers/nonsmokers)
gene: SPG21 | 217827_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3237
p=0.7891
p=0.0874
p=0.0955
p=0.143
p=0.7705
p=0.0109
p=0.1219
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

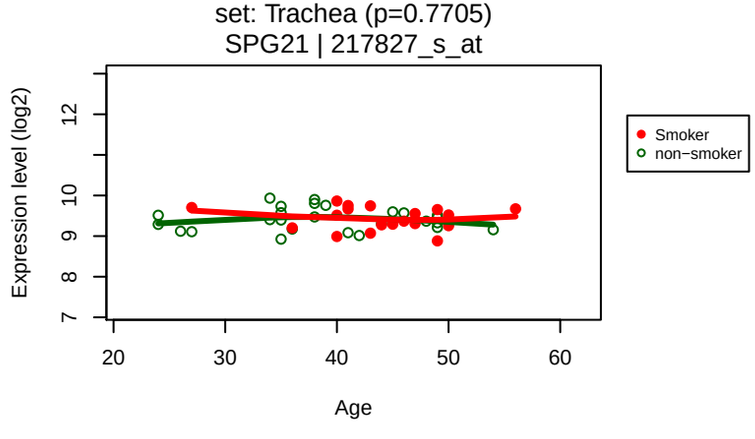

Gene expression ratio in Males
gene: SPG21 | 217827_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7426
p=0.0533
p=0.9294
p=0.0369
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: SPG21 | 217827_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7402
p=0.8438
p=0.8008
p=0.978
SmallAirway2
LargeAirway2
Trachea
Macrophage2

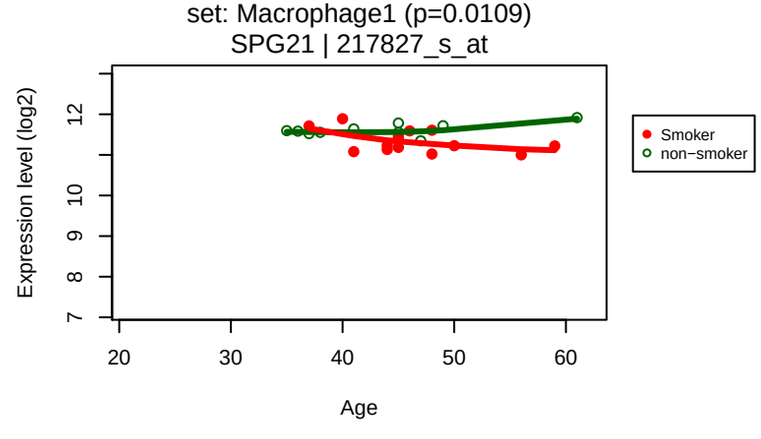

set: SmallAirway1 (p=0.3237)
SPG21 | 217827_s_at
Expression level (log2)
Age
Smoker
non−smoker

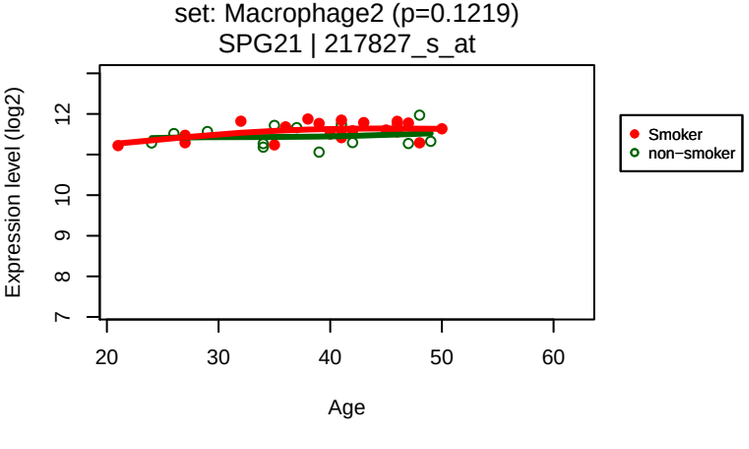

set: SmallAirway2 (p=0.7891)
SPG21 | 217827_s_at
Expression level (log2)
Age
Smoker
non−smoker

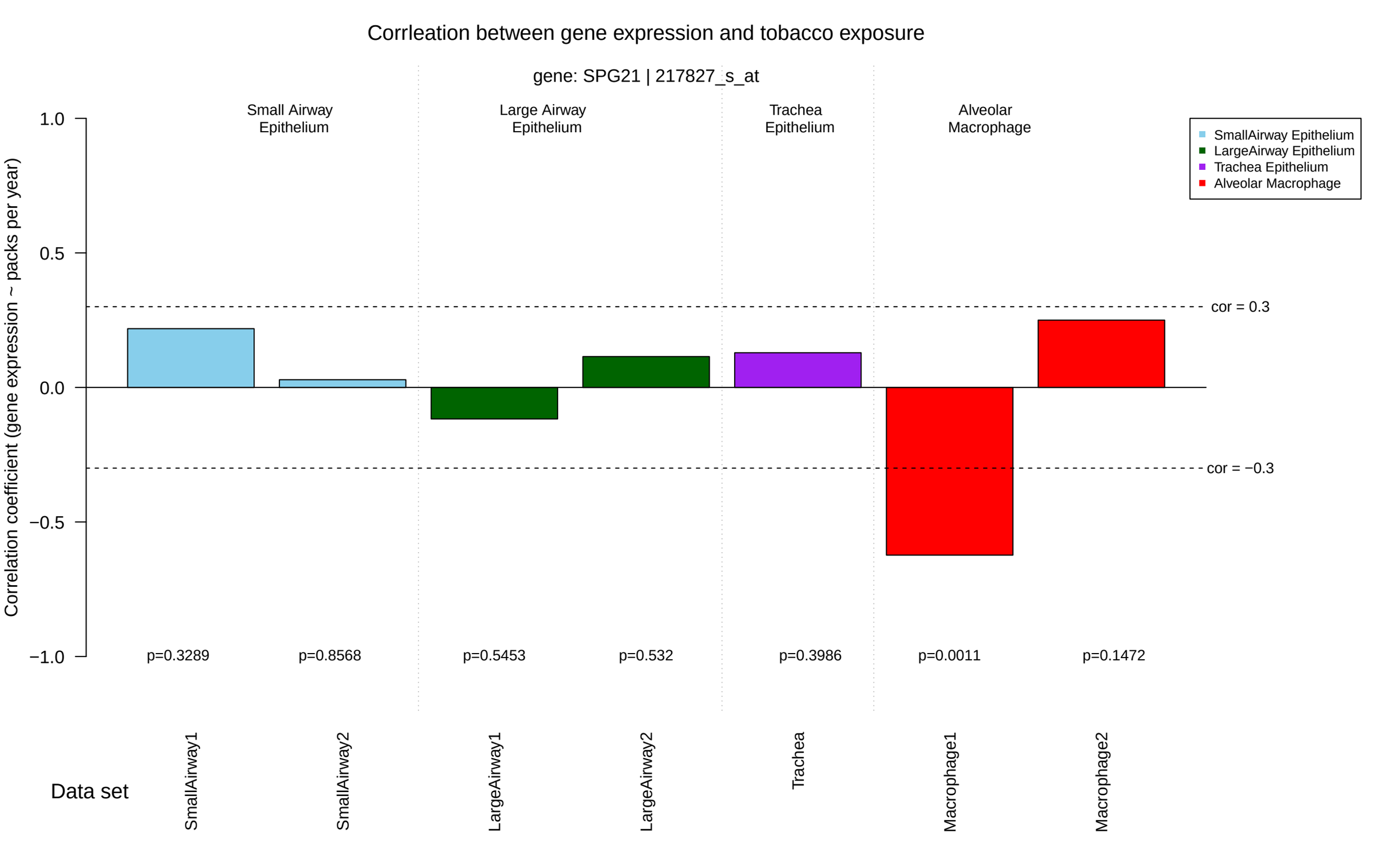

set: LargeAirway1 (p=0.0955)
SPG21 | 217827_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.143)
SPG21 | 217827_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.7705)
SPG21 | 217827_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0109)
SPG21 | 217827_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1219)
SPG21 | 217827_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: SPG21 | 217827_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.3289
p=0.8568
p=0.5453
p=0.532
p=0.3986
p=0.0011
p=0.1472
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2